

THE RESULTS

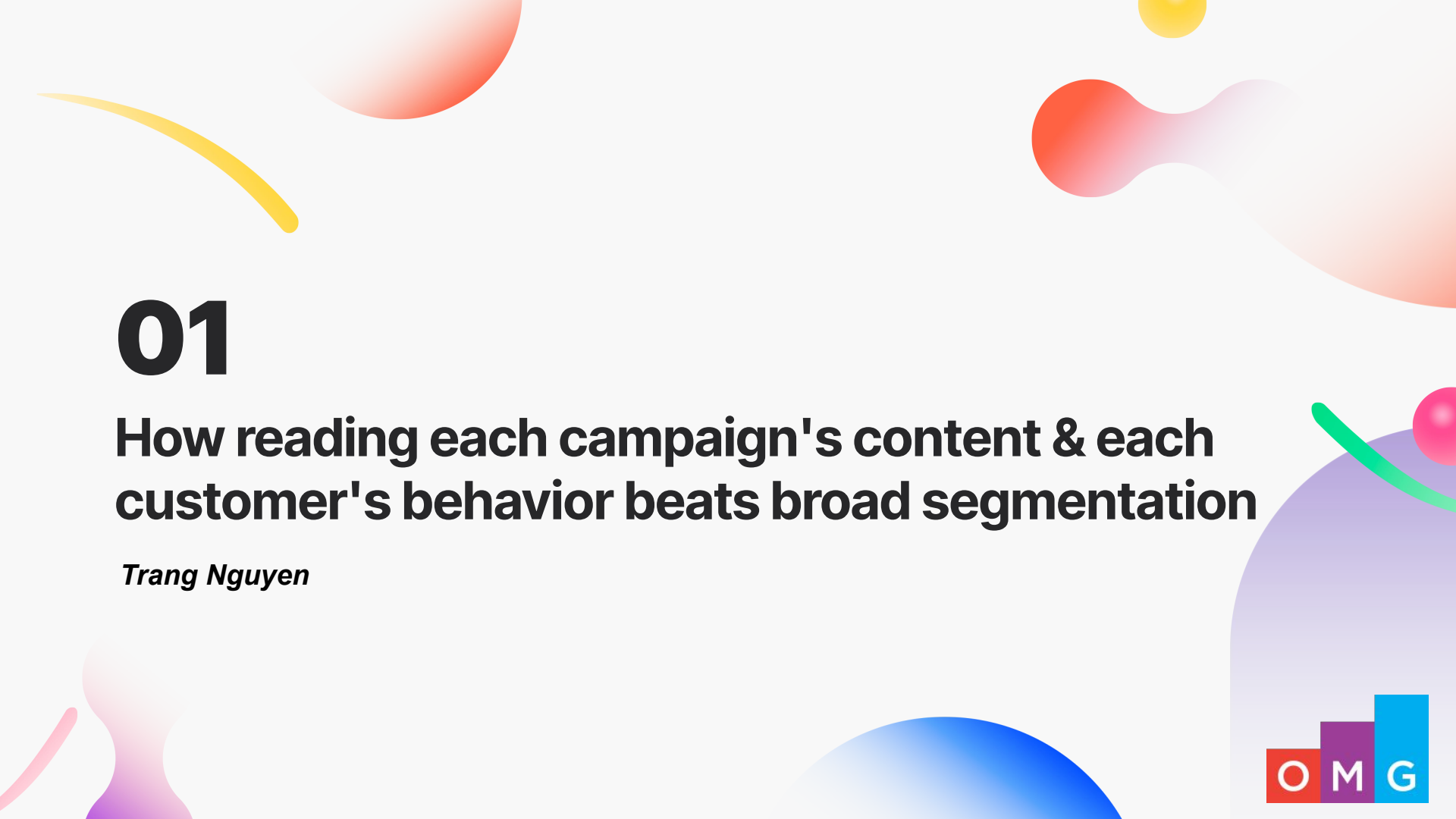
BSc Business Analytics Thesis

Company: *Online Marketing Group*

Supervisor: *Yunming Hui*



O M G



01

How reading each campaign's content & each customer's behavior beats broad segmentation

Trang Nguyen



The approach

01

Read every campaign's content

A Biterm Topic Model sorts each campaign's subject line into one of the content topics automatically, with no manual tagging.

02

Score each customer

A Decision Tree combines each user's past behavior with how well the content matches them, and predicts who will engage.

The result | one engagement score per customer, per campaign, produced automatically, at scale



STEP 1 - BITERM TOPIC MODEL

How the model finds a topic

A subject line is too short to read word-by-word, so the model learns from **biterns** : every pair of keywords. Pairs that keep co-occurring across campaigns form a theme.

1 · ONE SUBJECT LINE

“Een elektrische SUV voor de zakelijke rijder”

Only a handful of words, too few to read one at a time.

2 · KEEP KEYWORDS

elektrisch

SUV

zakelijke

rijder

Drop filler words like “een”, “voor”, “de”.

3 · MAKE WORD-PAIRS

rijder-elektrisch

elektrisch-SUV

rijder-SUV

zakelijke-SUV

elektrisch-zakelijke

rijder-zakelijke

4 keywords → **6 biterns** .
More signal from the same short text.

4 · POOL ACROSS ALL CAMPAIGNS

The top words for this theme:

auto

mazda

rijder

elektrisch

toyota

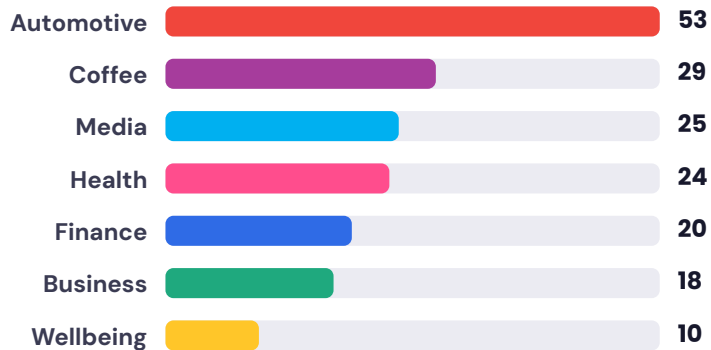
Labelled **Automotive & Mobility** · 53 campaigns

Topic classification

- The number of topics is determined manually



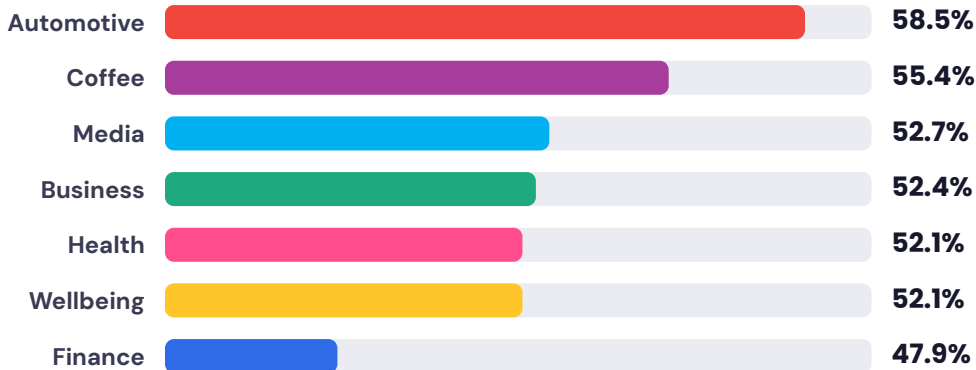
Campaigns per topic



Some content types get opened more than others

Open rate by content theme

share of emails opened



+10_{pts}

between the best and worst themes.

What you send shifts open rates as much as **who** you send it to.



STEP 2 - DECISION TREE

Scoring each customer

Each person is scored with a **decision tree**, a chain of simple yes/no questions, like a flowchart, that ends in a prediction



Behavior signals

How often they've opened & clicked before



Content-match signals

How well this campaign's theme fits them

STRONGEST SINGLE SIGNAL

Past open rate

If someone has opened before, they're far more likely to open again.



RESULT

Opens are habit, clicks are about content

THE OPEN

Driven by habit

Whether someone opens depends mostly on whether they've opened before, not on the topic.

~70% of the decision is past behavior

THE CLICK

Driven by content fit

Whether someone clicks depends much more on how well the campaign's theme matches their interests.

2× more weight on content than for opens

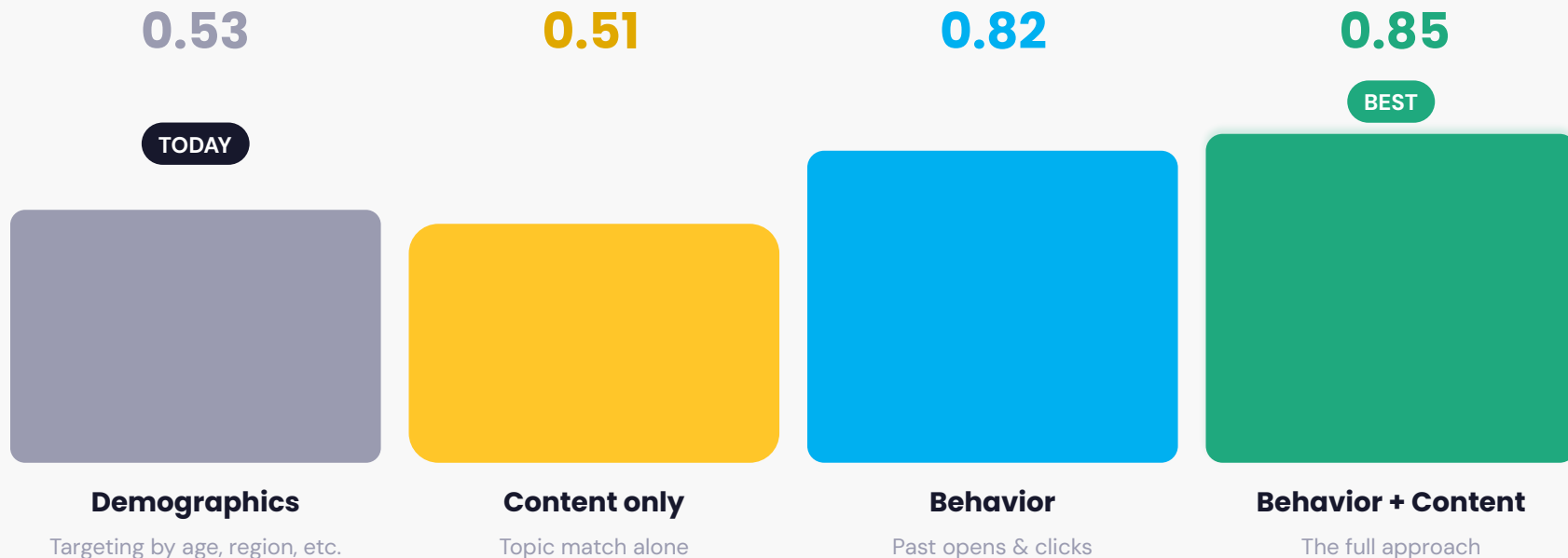


RESULT



The model beats how campaigns are targeted today

Accuracy of four targeting approaches (AUC: higher is better).



What you'd actually get: a ranked send list

Campaign targeting					
Campaign Automotive & Mobility "Een elektrische SUV voor de zakelijke rijder"					
RANK	PROFILE	PAST OPENS	CONTENT MATCH	ENGAGEMENT SCORE	ACTION
01	Profile #4821 Automotive · 41, M	82%	0.91	94	Send
02	Profile #1192 Automotive · 53, M	74%	0.88	88	Send
03	Profile #7705 Media · 36, F	69%	0.62	71	Send
04	Profile #3340 Finance · 29, F	38%	0.44	41	Hold
05	Profile #9014 Wellbeing · 47, M	22%	0.18	19	Hold

Recommendations

01

Rank recipients by content match before each send

Prioritize the users the campaign's theme fits best, especially for click-focused campaigns, where content matters most.

02

Lead with behavior, layer content on top, and retrain

Keep past behavior as the main signal, add content matching as a second, and refresh the model periodically as new responses arrive.



What to keep in mind



Modest sample

Built on 179 subject lines, and clicks were rare, so click findings are directional rather than precise.

Uneven content mix

Automotive made up ~70% of campaigns, so the seven themes aren't equally distributed in the data.

Clarity over raw accuracy

One readable decision tree is chosen for transparency, although a more complex model could lift accuracy further.

A larger, more balanced dataset would sharpen the exact numbers.



02 Predicting Email Engagement

Research question

To what extent do flexible tree-based models provide additional value over Logistic Regression?

- Predictive value
- Targeting value
- Explanatory value

Hien Nguyen

From campaign data to engagement decision support

Business problem

Email engagement differs across recipients and campaigns, but it is unclear which recipients are most likely to open or click

Artifact

A predictive analytics workflow that uses recipient, campaign/content, and historical engagement data to estimate open and click probabilities

Use

Supports campaign planning
Recipient prioritisation
Engagement analysis



How the models were evaluated

Prediction setup

Unit of analysis:

Recipient-campaign

Data:

1.16M observations

Outcomes:

Open and click

Model comparison

- **Logistic Regression (LR):**
Transparent benchmark
- **Random Forest (RF):**
Main flexible tree-based model

Evaluation design

- Grouped stratified cross-validation for development
- Chronological holdout test for final evaluation

Random Forest did not universally dominate

Task	Key result	Interpretation
Open prediction	RF: slightly higher ROC-AUC and PR-AUC LR: better Brier score and log loss	RF ranked recipients slightly better LR gave better probability quality
Click prediction	PR-AUC: 0.126 (LR) vs 0.122 (RF) ROC-AUC: 0.968 (LR) vs 0.971 (RF)	LR slightly led on primary rare-event metric RF slightly led on broad ranking
Overall	No consistent winner across metrics	Complexity alone did not guarantee better performance

Random Forest added value in narrow click targeting

Final test set: 195 clicks

Top-ranked segment	Recipient selected	Logistic Regression	Random Forest
Top 0.1%	290	32 clicks	40 clicks
Top 0.5%	1450	103 clicks	117 clicks
Top 1%	2890	153 clicks	161 clicks

Random Forest's clearest advantage appeared at the very top of the click ranking



Random Forest did not reveal a completely different explanation

Historical engagement

Dominant predictor group across models

Campaign/content features

Secondary but useful signal

Recipient characteristics

Useful, but less dominant

Random Forest

Relied on similar broad signals as Logistic Regression, but represented them more flexibly
Added flexibility, but not a fundamentally different explanation of engagement



Use as decision support, not automatic decision-making

Reliability limits

- Click events were extremely rare
- Final-test performance was lower than cross-validation
- Interpretation is descriptive, not causal

Before operational use

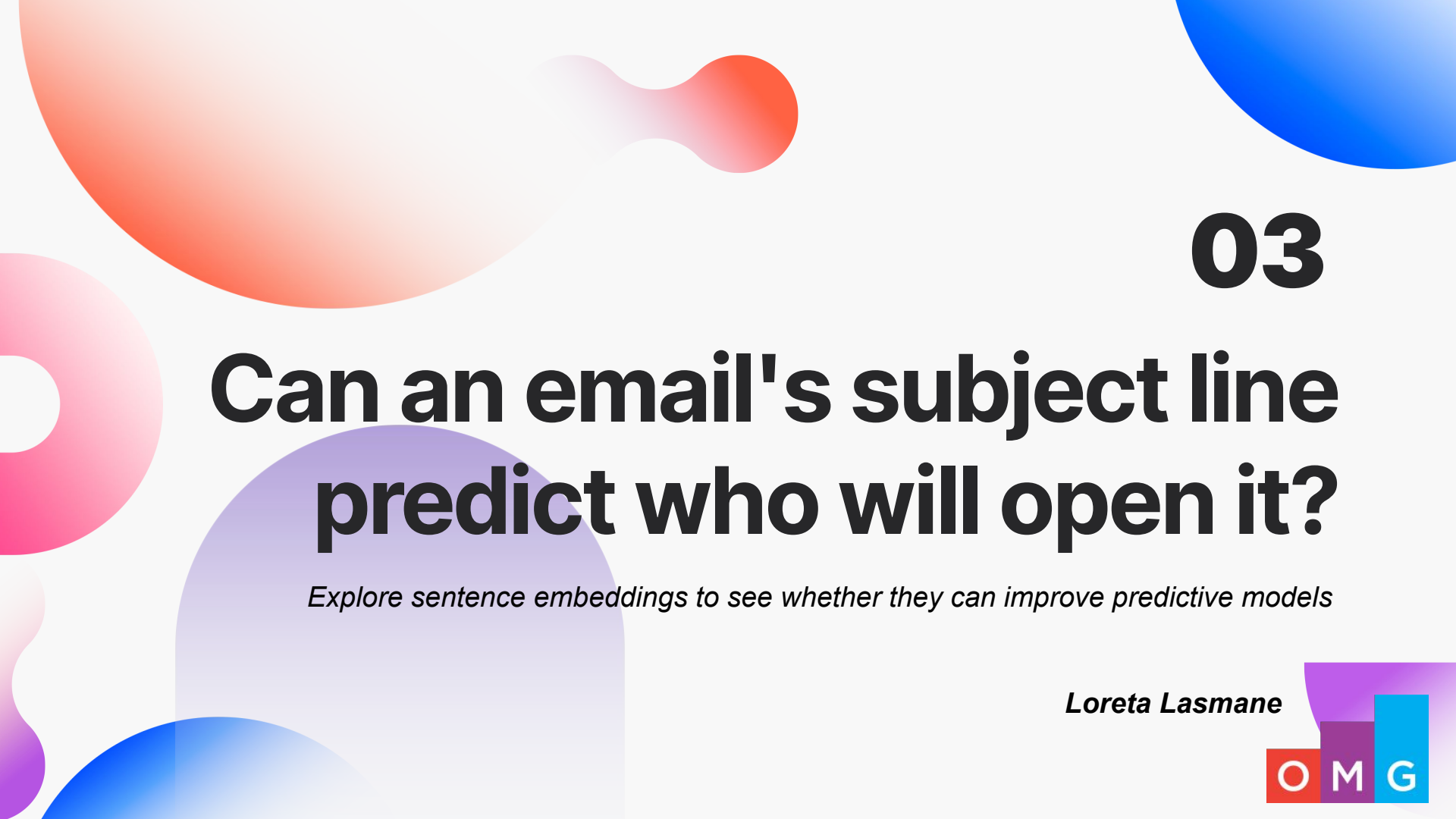
- Validate on future campaigns
- Monitor performance over time
- Handle recipient data responsibly

Final takeaway

Model choice should depend on the use case

- Logistic Regression for broad prediction and explanation
- Random Forest for narrow click prioritisation.





03

Can an email's subject line predict who will open it?

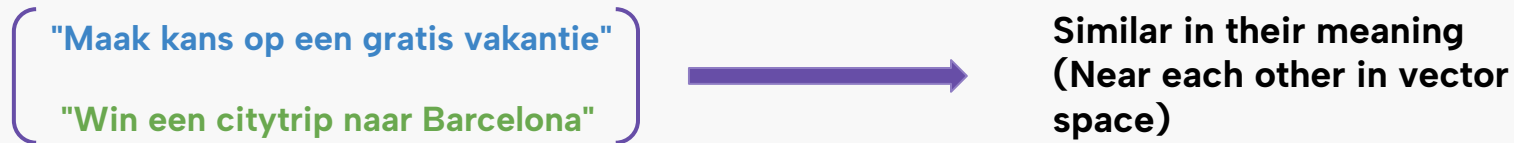
Explore sentence embeddings to see whether they can improve predictive models

Loreta Lasmane



Main idea

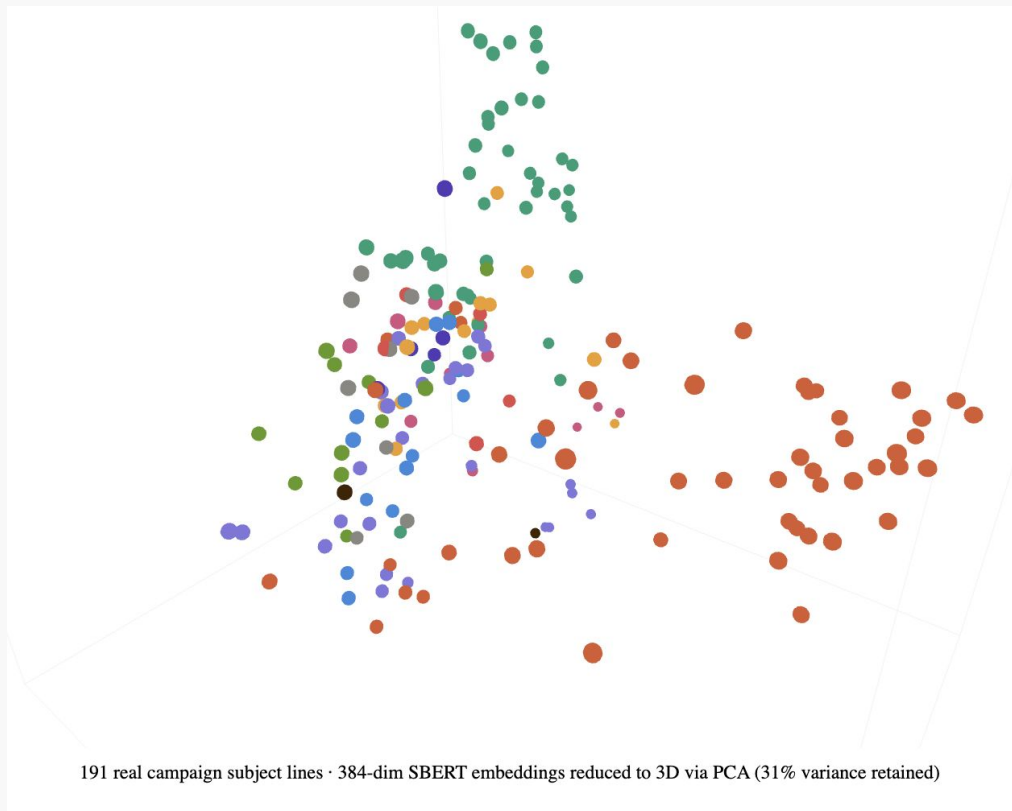
- OMG currently targets by broad segment, but two customers in the same segment can have completely different content preferences
- **Deriving text-based features from e-mail subject lines to see whether they improve predictive models of individual open probability**



Subject lines as **word embeddings**?

A word embedding is a way of representing a word as a list of numbers (a vector), positioned in space so that words with similar meaning end up close together, even if they don't share any letters or spelling

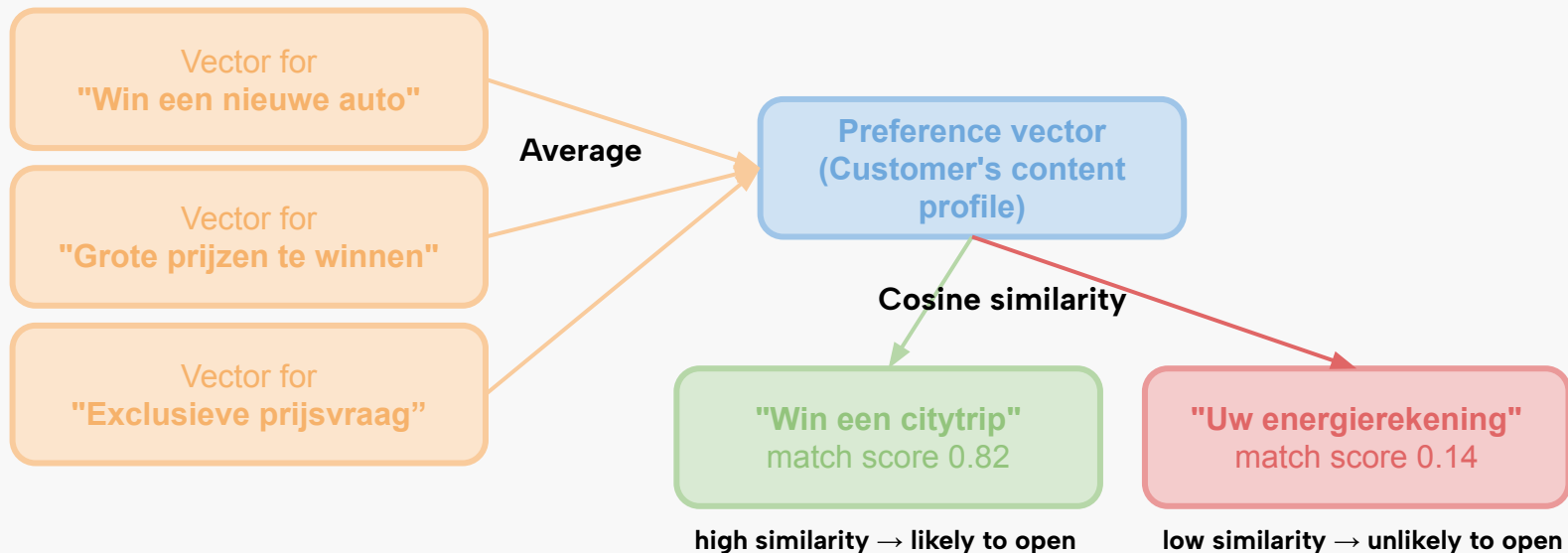
Each dot is a real OMG email. Similar emails sit close together, even without sharing a single word



How it works?

Build a 'taste profile' for each customer, then check how well a new email matches it.

Previously opened e-mails by a customer



Two subject line features derived:

1. Semantic Match Score

- Measures **how well a new email's subject line matches what a specific customer has opened before.**
- Built a "**taste profile**" for each customer based on their past opens, then check how closely a new email lines up with that profile.

2. Novelty Score

- Measures **how different a new email is from what a customer usually receives**, regardless of whether they've opened those past emails or not.

Results

- Behavioral history alone predicts opens well (AUC 0.80)
- Adding subject line embeddings to a simple model showed no improvement
- Adding the same embeddings to a non-linear model did improve accuracy (AUC 0.81) and the **semantic match score** became the **2nd most predictive feature overall**, just behind past open rate

Model	Method	Features	Mean AUC	± Std
M1	Logistic Regression	Past open rate only	0.8050	0.0151
M2	Logistic Regression	Behavioral + Demographic	0.8002	0.0233
M3	Logistic Regression	Behavioral + Demographic + Semantic	0.7879	0.0187
M4	Random Forest	Behavioral + Demographic + Semantic	0.8108	0.0231

Model Performance Comparison 5-Fold Cross-Validation (AUC-ROC)

Results: Semantic Feature Coefficients

M3 Logistic Regression coefficients:

- Both semantic features have by far the largest coefficients in the entire model (+3.61 and +2.74), bigger than past_open_rate itself (+1.09)

M4 Random Forest feature importance:

- Semantic match score is the 2nd most important feature out of all of them, passing every behavioral signal except past open rate itself
- Novelty score also ranks above every topic category, coming in as the 4th most important feature overall. This means how different an email feels from a customer's usual inbox carries real predictive value on its own, not just how well it matches their taste.

Takeaway:

- The two semantic features turned out to be one of the strongest individual predictors in models
- They revealed real patterns in customer behavior that weren't visible using behavioral data alone, but because the two signals overlap so closely with each other, they didn't directly translate into a more accurate simple model; their value only showed up once we used a model "smart" enough to handle that overlap.



Recommendations

Key takeaways:

- The system builds a "taste profile" for each customer automatically, just from their open history
- **Past behavior is still your strongest signal**, keep prioritizing engaged customers first
- Content-based matching adds real value once paired with a model that can use it properly

Before building a live dashboard, consider:

- **More variety in content** would sharpen the signal further
- This **works automatically on new subject lines**, no manual tagging needed
- Currently scored at training time, not real-time at the moment of send
- **Needs periodic retraining as customer behavior** and campaign content evolve
- **Best used to support targeting decisions**, not replace human judgment entirely

What this could look like?

Campaign send recommender

Score a new email against your customer base

↻ New scoring

Subject line

Win een citytrip naar Barcelona

Preheader (optional)

Doe nu mee en maak kans op een onvergetelijke citytrip

✦ Score customers

Save as draft

Customers scored

20,000

Recommended sends

5,420

Avg open probability

68%

Avg semantic match

0.71

Ranked customers

Send to top 27%

Customer	Open probability	Match score	Novelty score	Past open rate	Decision
user_04821	91%	0.88	0.34	74%	<button>Send</button>
user_11932	86%	0.81	0.52	69%	<button>Send</button>
user_00557	79%	0.72	0.61	58%	<button>Send</button>

04

Do email engagement models improve as they see more data?

Using sequential batch retraining to investigate whether periodic model updates improve email engagement prediction over time.

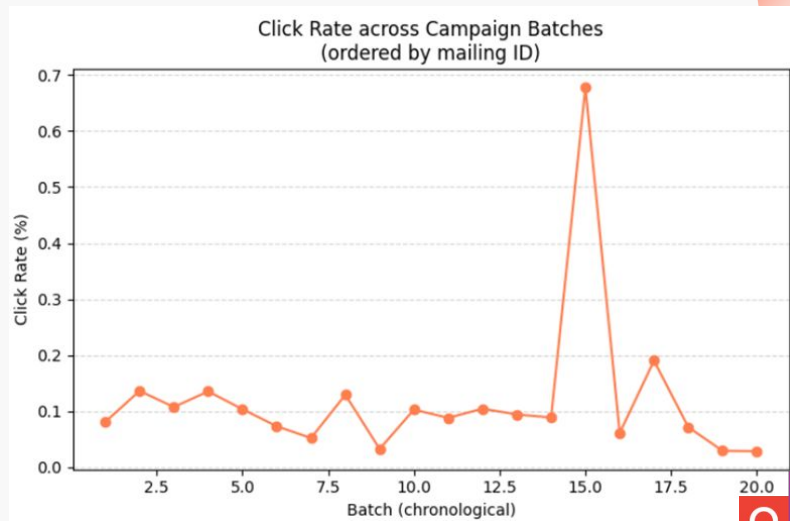
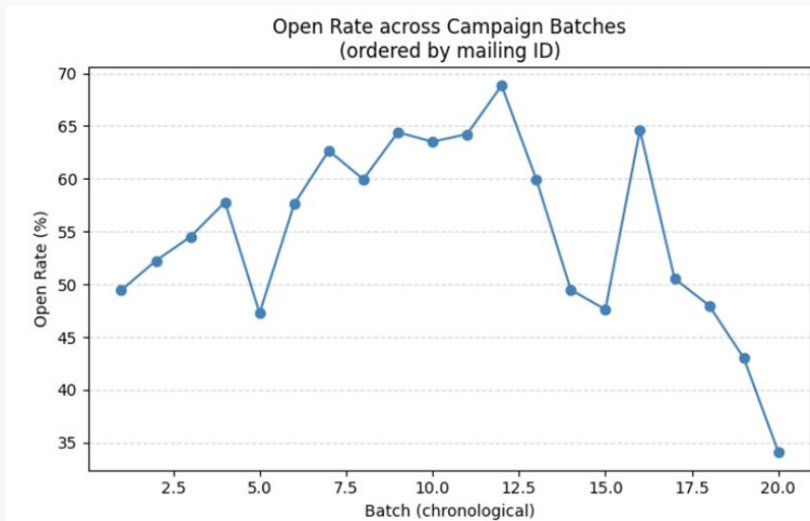
Giada Frontini



Email engagement shifts over time

Temporal stability analysis of engagement rates

- Open rates vary across campaign batches (34.1% – 68.9%, SD = 0.088)
- Click rates are mostly stable but with an anomalous spike at batch 15 (0.68% vs. 0.03–0.13% of surrounding batches), likely a campaign specific effect

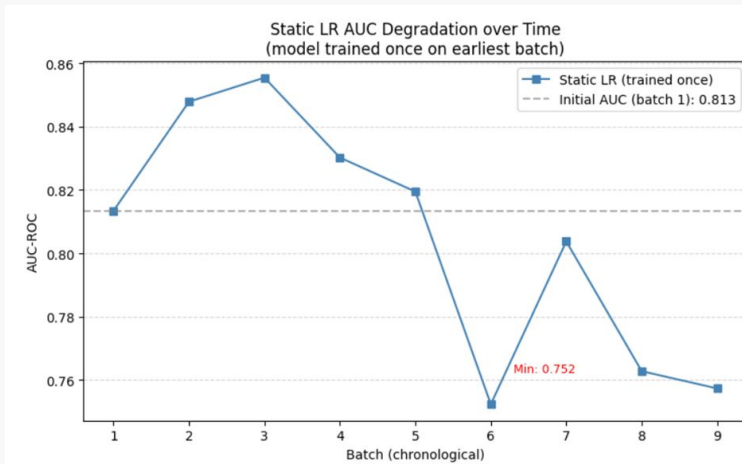


Email engagement shifts over time

Diagnostic check: logistic regression trained once on the first batch and evaluated on each subsequent batch without updating

- AUC declined from 0.813 to 0.752 over 9 batches (losing 7.5 percentage points)
- This is consistent with concept drift, as the gap between training and test data grows, the model's ability to generalise declines

→ This motivates the investigation of whether periodic retraining can maintain performance



The adaptive retraining framework

Expanding and sliding window strategies evaluated against a static baseline

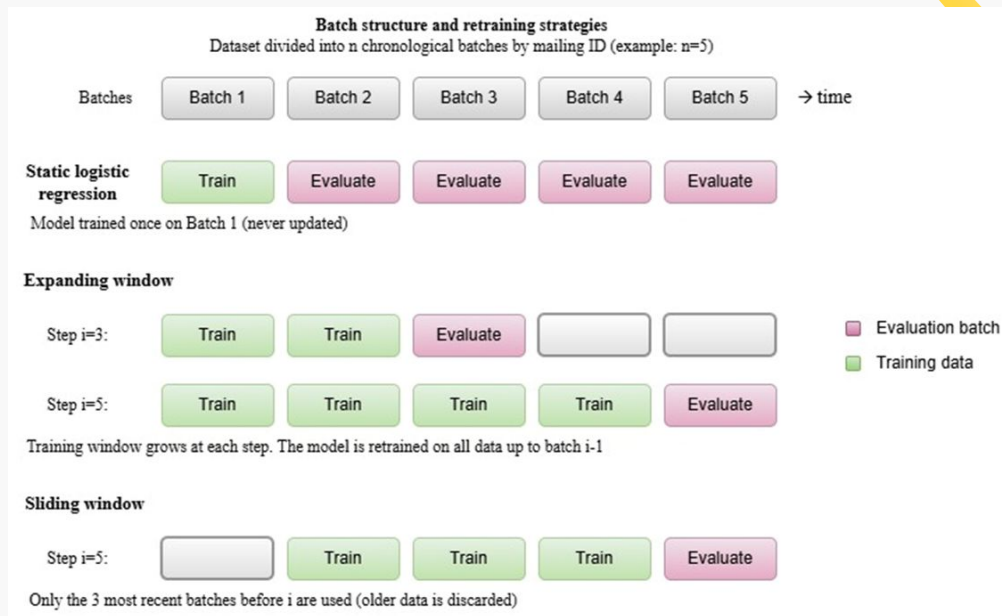
Dataset is divided into n chronological batches using mailing ID as a chronological proxy

Expanding window: model retrained on all data up to batch i , evaluated on batch i

Sliding window: model retrained on the 3 most recent batches only, discarding older data

Primary configuration: $n=10$ batches (approximately ~27 mailings, ~100,000 observations per batch)

At each step: retrain → evaluate on next batch → compare to static model

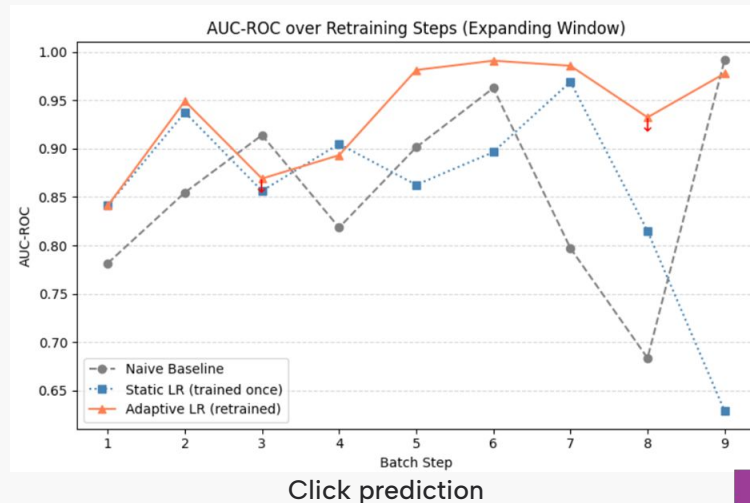
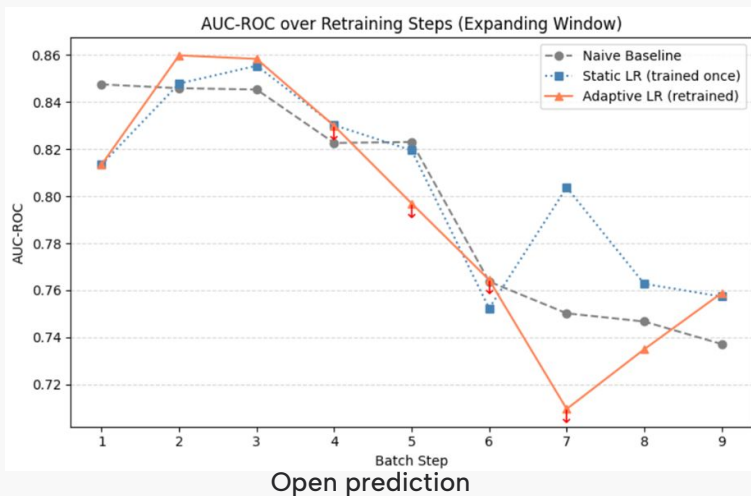


Results: primary configuration (n=10)

Open prediction: adaptive mean AUC 0.7918 vs. static 0.8047. No significant difference (Wilcoxon $p = 0.641$)

Click prediction: adaptive mean AUC 0.9355 vs. static 0.8568, winning 7/9 batches. Significant improvement (Wilcoxon $p = 0.023$)

Sliding window was also evaluated for both targets and results consistent with expanding window findings.

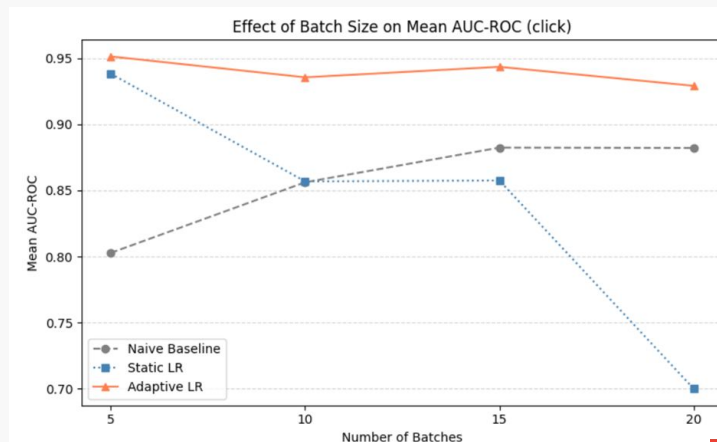
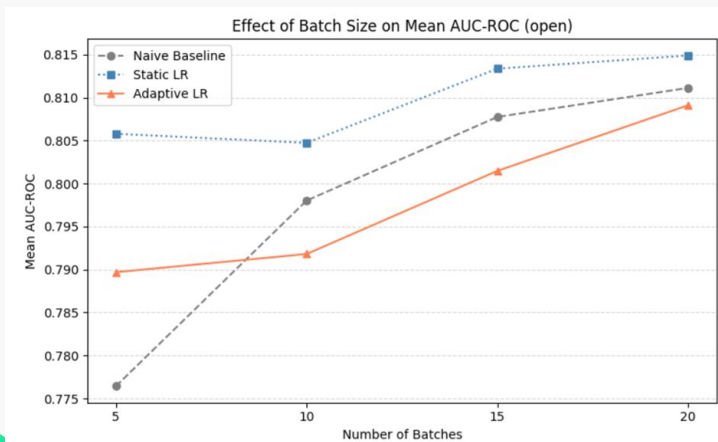


Batch size sensitivity analysis

Expanding window evaluated across $n = 5, 10, 15, 20$ batches

- Open prediction: adaptive model win rate increases with n , from 0/4 at $n=5$ to 13/19 at $n=20$. Mean AUC improves for all models as n increases
- Click prediction: adaptive mean AUC remains high across all batch sizes. Static model declines at $n=20$ (mean AUC 0.700), partly due to extremely limited training data at that configuration

→ $n=20$ is selected as best performing configuration by win rate criterion for both targets

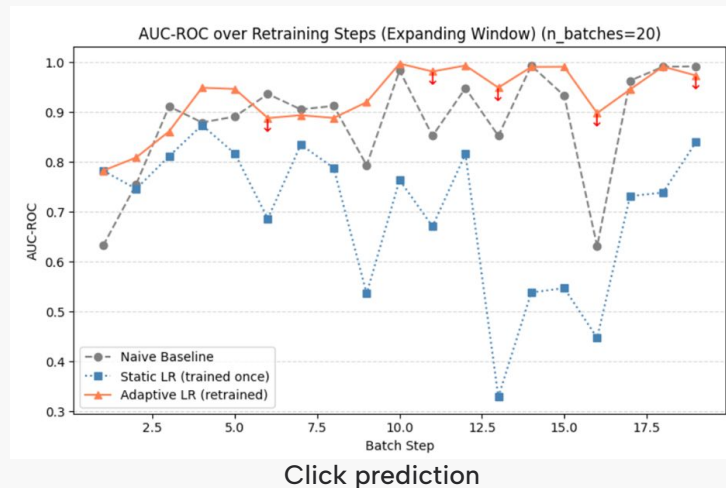
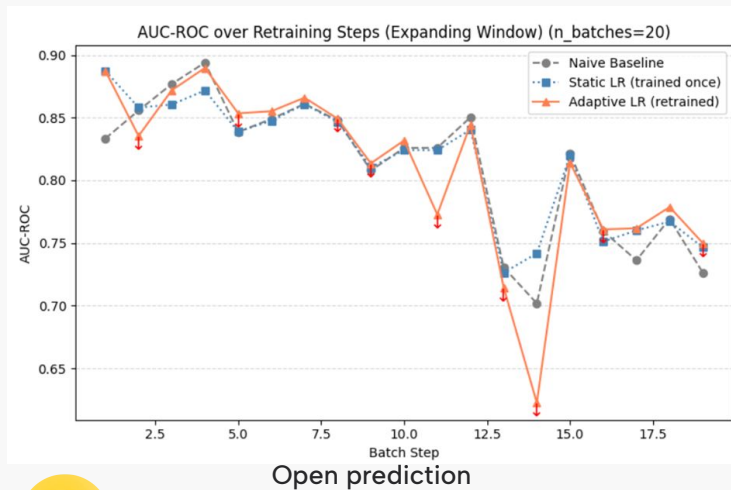


Results: optimal configuration (n=20)

Open prediction: adaptive mean AUC 0.8091 vs. static 0.8149. No significant difference (Wilcoxon $p = 0.5277$), consistent with $n=10$ finding

Click prediction: adaptive mean AUC 0.9291 vs. static 0.7000, winning in 18/19 batches. Highly significant improvement (Wilcoxon $p = 0.0002$)

A notable AUC drop is observed at batch 14 (AUC = 0.623 for the adaptive model) in the open prediction, consistent with the drop at batch 7 in the $n=10$ configuration. Both correspond to the same period of low engagement in the data.





Date-based batching

Open prediction. Expanding window repeated using actual send dates (168/274 mailings with temporal data)

- Results consistent with primary mailing ID based analysis. Static outperforms adaptive for open prediction (mean AUC 0.8246 vs. 0.8229)
- Confirms that the lack of significant improvement from retraining for open prediction is not an artefact of using mailing ID as a chronological proxy

→ Primary findings are robust to the choice of batching scheme

Model	ID based split	Date based split
Naive baseline	0.7980	0.8088
Static logistic regression	0.8047	0.8246
Adaptive logistic regression	0.7918	0.8229

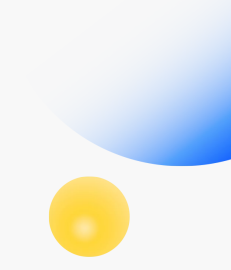
Conclusions

The value of retraining is target dependent

- Open prediction: periodic retraining does not significantly outperform a static model. Static logistic regression is sufficient. Monitor periodically for drift
- Click prediction: periodic retraining produces a significant AUC improvement across all batch sizes tested. Note that severe class imbalance (0.1% positive rate) means results should be interpreted with caution.

Design principle: retraining frequency should be allocated based on empirical evidence of performance degradation per prediction target, not applied uniformly.

Recommendation: retrain the click model periodically, keep the open model static and monitor for drift



05

Designing a Personalised Email Recommendation System

Using machine learning to predict engagement and support personalized campaign targeting

Maia Villar



Methodology

STEP 1

Data Collection and Preparation

Combine user engagement data with campaign metadata.

STEP 2

Exploratory Data Analysis (EDA)

Exploratory analysis to identify factors influencing opens and clicks

STEP 3

Feature Engineering

Convert raw data into meaningful features for engagement prediction.

STEP 4

Predictive Modelling

Train and compare ML models to predict engagement probabilities: Logistic Regression, Random Forest and XGBoost.

STEP 5

Recommendation System

Rank users based on predicted engagement to support campaign targeting.



What drives engagement?

Key Finding #1

Past Behavior Predicts Future Behavior

Users who frequently opened and clicked emails in the past are significantly more likely to engage again.

Strongest predictors across models:

- Past Open Rate
- Past Click Rate
- Past Opens/Clicks

Key Finding #2

Recent Behavior Matters Even More

Recent engagement is a stronger predictor than long-term averages.

Behavioral Recency Variables:

- Open Rate in Last 5 Campaigns
- Click Rate in Last 5 Campaigns

Impact on model performance:

- Open ROC-AUC: 0.824 → 0.846
- Largest improvement of all feature engineering additions

Key Finding #3

Email Fatigue Exists

Users who receive more emails become less likely to engage.

- Higher email exposure → lower open probability
- Higher email exposure → lower click probability

Model Comparison

Performance Evaluation (ROC-AUC)

Model	Open Prediction	Click Prediction
Logistic Regression	0.824	0.968
+ Behavioral Recency	0.846	0.969
+ Topic Affinity	0.848	0.970
Random Forest	0.869	0.988
XGBoost	0.875	0.984

**Best Open
Model**

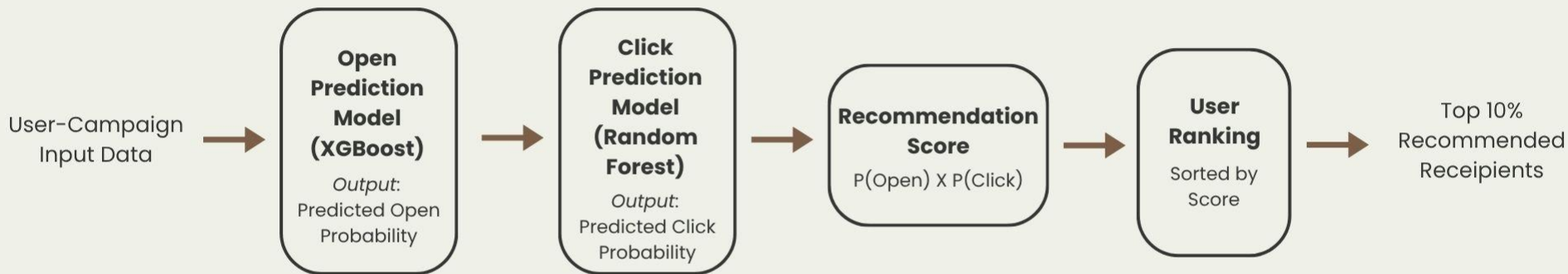
XGBoost

**Best Click
Model**

Random Forest

The best results were achieved by combining *recent user behavior*, *content preferences*, and advanced machine learning models.

Recommender System



Example Recommender System Output:

user_id	mailing_id	subject_line	main_topic	open_probability	click_probability	recommendation_score
5c6bed415798a7942836793a	880	Rijd dit jaar nog onder het gunstige bijtellin...	Automotive & Mobility	0.918940	0.981910	0.902317
5cb9a38736dd1336b9c558f8	1107	Is dit jouw volgende zakelijke auto?	Automotive & Mobility	0.934121	0.965315	0.901721
5c6bed415798a7942836793a	909	Elke dag een goed begin? Ontdek Nespresso Prof...	Energy & Sustainability	0.943536	0.954277	0.900394

Recommendation Results

Because a live implementation was outside the scope of this study, recommendation quality was evaluated using **predicted engagement probabilities** rather than observed campaign outcomes.



Business Implications for OMG



1. **Prioritize Recently Engaged Users**

Use recent opens and clicks as key targeting signals.



2. **Reduce Email Fatigue**

Send fewer, more relevant emails rather than maximizing volume.



3. **Deliver More Relevant Content**

Match campaign content to users' demonstrated interests.



4. **Move Toward Profile-Level Targeting**

Complement traditional segmentation with personalized recommendations.

The findings suggest that OMG can **improve engagement** by combining behavioral data, content preferences, and machine learning-driven recommendations to support more personalized campaign decisions.

Limitations and Next Steps

Limitations:

1

Offline evaluation only

- Recommendations were evaluated using historical data.
- No live deployment was conducted.

2

Predictions, not outcomes

- Results are based on predicted engagement probabilities.
- Actual engagement impact remains to be validated.

3

Limited Feature Availability

- Only available profile, behavioral, and campaign variables were used.
- Additional contextual data could further improve performance.

Next Steps

Conduct a live A/B test

- Update recommendations as new campaign data becomes available.

Expand personalization

- Compare model-driven targeting against the current approach.
- Measure open, click, and conversion rates.

Implement continuous model retraining

- Incorporate additional content and contextual features.
- Explore send-time optimization at the individual level.

Thank you for your attention!

Giada Frontini, Loreta Lasmame, Hien Nguyen,

Trang Nguyen, Maia Villar

